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RESEARCH ARTICLE

Continuous Speech-Based Fatigue Detection and Transition State Prediction for Air Traffic Controllers

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ABSTRACT Air traffic controllers (ATC) play a critical role in ensuring aviation safety, but their demanding workload can lead to fatigue, potentially compromising their performance. This paper presents a study that investigates speech features responsible for detecting ATC fatigue and proposes an approach to predict the timestamp at which an ATC transitions into a fatigue state from a continuous speech sample. The main contributions of this work are the creation of a continuous speech ATC dataset and the identification of a lightweight optimum feature set for fatigue classification from ATC speech. For the initial task, the classification of raw speech signals into fatigue and non-fatigue categories was performed using the top-10 best features selected from the openSMILE feature set. The evaluation was carried out using various learning algorithms such as XGBoost, Adaboost, Random Forest, HistogramGB, and 1D-CNN. The ensemble algorithms demonstrated the best performance, achieving a maximum accuracy of 100% on the XGBoost test set. Further, interpretability was analyzed using the SHAP tool, which identified the prominent features for the task. The second task involved creating a continuous speech dataset comprising approximately 18,900 samples from the ATC corpus, with an average duration of 63-65 seconds per sample. The continuous speech samples were prepared by the randomized concatenation of fatigue and non-fatigue chunks, each with a duration of approximately 15 seconds. Automated sequence labeling was performed on uniformly segmented continuous speech samples. MFCC and statistical features were extracted from the labelled continuous speech and input into various Recurrent Neural Networks, such as bi-LSTM and Bi-GRU, for fatigue state prediction tasks. A combination of these features using bi-LSTM modeling achieved a maximum precision, recall, F-score, and average accuracy of 99% each. Finally, sample-wise timestamp prediction was performed using the labels: fatigue, non-fatigue, and ambiguous (transition). To the best of the authors' knowledge, this research is the first of its kind to address continuous speech-based fatigue state prediction for ATCs. All tasks were conducted using the Civil Aviation Administration of China ATC corpus.

INDEX TERMS Air traffic controller, continuous speech, XGBoost, recurrent neural networks, bi-LSTM, MFCC.

I. INTRODUCTION

According to a 'Times of India' news article survey [1], more than 66% of pilots reported falling asleep in the cockpit or experiencing micro-sleep episodes while working, due to

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weariness and ''daytime fatigue/sleepiness''. Additionally, a survey conducted by NGO Safety Matters Foundation showed that approximately 54% of the pilots experience higher instances of daytime sleepiness, while 41% of pilots suffer from mild drowsiness [2]. Fatigue can have a serious impact on Air Traffic Controllers (ATCs) who are responsible for the safe flow of air traffic. They work in various environments, including air traffic control towers and radar approach control centres. This role demands a high workload, long work shifts, quick reaction time and accurate decision-making ability. Fatigue resulting from these factors can lead to errors in judgment and performance, which can have disastrous consequences. Studies [3] have shown a strong connection exists between increased weariness and safety risks. Moreover, fatigue is a poorly defined concept, as it is a complex and multifaceted issue with various underlying causes, especially biological and psychological. Given this complexity, researchers in the air transport field have placed significant focus on real-time detection of ATC fatigue. The speech-based fatigue detection approach, known for its high accuracy and adaptability, has garnered considerable attention in recent research. In both international aviation and military air forces, drowsiness presents a major safety concern, particularly in continuous speech tasks. Symptoms due to prolonged state of being in fatigue have been investigated in [4], leading to health concerns including sore throat, difficulty in thinking and migraine attacks. While complete prevention of fatigue may not be feasible, recognizing and addressing it through necessary actions can certainly improve the physical well-being of ATCs [5].

A study [6] analyzing fatigue incidents among UK pilots to determine the level of tiredness a pilot can experience while still ensuring a safe flight. The implications of their research methods were only discussed in the context of aviation safety standards. Multiple approaches have been implemented for tackling fatigue by placing a camera to capture human behaviour. The images captured by the imaging systems are highly sensitive to interference caused by factors such as the camera's positioning, ventilation, dust, and other environmental conditions. Hence, detecting weariness based on assessing changes in the properties of speech signals offers some distinct advantages. Vocal-feature-based approaches have recently been the popular method for study due to the rapid advancements in voice recognition. Because ATCs' primary duty is to communicate with pilots by radiotelephone and laws require that all call records be kept for a certain amount of time, vocal features are easy to gather and analyze. Audio data can be generated during common procedures from the voice (speech) response of pilots. These voice signals contain a wealth of physiological data related to human exhaustion that can be utilized to gauge the level of pilot fatigue. The voice recognition system also has additional benefits like easy data collection, affordable hardware, straightforward operation, advanced de-noising techniques, excellent environmental adaptation, non-invasive, real-time, and so on.

Most of the existing methods for voice fatigue categorization have mainly been dependent on features like mel frequency cepstral coefficients (MFCC) and their extraction [7], [8], [9]. Audio descriptors may contain data on the emotive and physiological cues in a speech sample, important for disease detection and mental state analysis [10], [11]. For such emotional data, a class-specific latent affective structure model (LASM) is proposed in [12] to model the relationships among the emotion codewords for mood disorder detection which outperformed the traditional classifiers. Another recent approach [13] was using recurrent autoencoders on mel-spectrograms generated from raw speech input followed by support vector regressor-based predictions. This further led to advancements in methods to assess the significance of changes in prosodic parameters, such as fundamental frequency (F0) or sound pressure level, across extended periods. The likelihood of pilots experiencing severe weariness is maximum when the duty period is totally at the trough of circadian pattern/rhythm.

Recently, authors in [14] proposed a unique deep learning framework using the ATC fatigue dataset to counteract the loss of information by integrating active learning (AL) into complicated speech features chosen from a huge pool of unlabeled voice data. They developed a novel hierarchical stacked sparse autoencoder (SSAE) and densely connected convolutional automatic encoder (DCAE) network incorporated with AL. While the existing literature has observed alterations in explicit voice parameters such as fundamental frequency and word duration, no single voice characteristic consistently and reliably changes when speakers experience fatigue. Moreover, most research uses an expanded feature set for fatigue detection with discrete speech samples. To the best of our knowledge, no prior work has explored the use of a reduced set of optimized features designed specifically for efficient hardware implementation. Additionally, approaches to signify the fatigue variations in continuous speech for ATC perspective are much less explored. Our research focuses on establishing a connection between discrete and continuous aspects of speech in fatigue conditions for ATCs. The main contributions of this research are listed below:

- **Light-weight Feature set based Discrete speech classification**: Fatigue and non-fatigue classes from the ATC radiotelephony corpus are classified using a lightweight model trained on the top 10 features, identified through feature selection with multiple data balancing techniques and verified using explainable AI tools, followed by evaluation across various learning algorithms.
- **Continuous speech dataset creation & Automated sequence labelling:** A data corpus of 18,900 continuous speech samples, each with an average duration of over a minute is created through randomised selection and concatenation of discrete speech segments of fatigue and non-fatigue chunks. The continuous data created is uniformly segmented and subjected to automatic sequence labelling and appropriate tags viz. F (fatigue),

NF (non-fatigue) and A (ambiguous) referring to the transition state from F to NF or vice-versa) are provided.
- **Predictive modelling and fatigue transition timestamp prediction:** Prediction of transition state from non-fatigue to fatigue or vice versa as well as the duration and timestamps of fatigue and non-fatigue states, using recurrent neural network models.

The rest of the paper is structured as follows - Section II describes the detailed literature survey in the domain. Section III details the entire methodology in fatigue classification, continuous speech sample creation and finally timestamp prediction of state transition from fatigue to non-fatigue or vice versa. Section IV features the brief results including the analysis and performance evaluation of different classifiers and sequence models. Finally, we provide useful insights and conclude this research work in Section V.

II. BACKGROUND

The research in voice fatigue classification started as early as 1998. Several researchers have contributed to the detection of fatigue using various modalities ranging from speech, image and physiological signals. Researchers have proposed various studies to extract a wide range of features from voice recordings, including acoustic features, phonetic features, and physiological features, with details as discussed in the following subsections.

A. EARLY STUDIES

Buekers in [15] describes about Voice Interval Test (VIT) by comparison of voice performances in 20 patients with a history of vocal fatigue with that in 12 healthy subjects. The VIT is a voice endurance test that involves reading aloud for a long time often used to evaluate vocal fatigue. The subject's voice is recorded and analyzed for changes in acoustic features such as fundamental frequency (F0), sound pressure level (SPL), and jitter. Buekers found that the VIT was not able to distinguish between the patients with vocal fatigue and the healthy subjects. Following this, other studies have also found that voice endurance tests are not reliable measures of vocal fatigue. Laukkanen et al. [16] investigated the relationship between self-reported vocal fatigue and acoustic measures of voice quality after a voice endurance task involving 79 female primary school teachers. A more sophisticated set of experiments was conducted in [17], wherein they found that many acoustic features were altered in the lab experiment, but only a few were altered in the real-world experiment. This suggests that voice analysis may be a useful tool for detecting pilot fatigue in the lab, but more research is needed to see if it is effective in real-world settings. In [18], Li et al. introduced an advanced method for driver fatigue detection, combining fuzzy support vector machines (SVMs) with nonlinear audio analysis. They focused on nonlinear dynamic properties like fractal dimension, and estimated entropy, reconstructing the voice signal in a high-dimensional space to create a robust nonlinear model. Addressing issues related to loudness, noise, and outliers, they integrated fuzzy cluster analysis into the SVM classifier during the design of a multi-feature fatigue classification model. The proposed model demonstrated a remarkable overall accuracy of 91.2%. A comparative study for vocal disorder identification was conducted in [8] and involved three classes, including healthy participants and two sets of distinct disorders utilizing MFCC and classifiers such as SVM and Gaussian Mixture Models (GMM). The GMM approach exhibited a 74% accuracy for continuous speech, while the SVM system achieved 72% accuracy in their comprehensive evaluation. In [9], the significance of precise phonetic alignment and categorization for enhancing speech-based fatigue detection was demonstrated. A technique for sports exhaustion identification was devised using feature extraction and an SVM-based machine learning system. The study included 15 recordings of women and men with English as their first language for training and testing. After meticulous tuning and evaluation, their optimal model achieved an impressive accuracy of 91.67%.

During the initial research period in this domain, researchers mainly employed simple software tools for identifying and classifying voice fatigue by analyzing features like fundamental frequency, short-time energy, and power spectral density to understand the nuances of speech about fatigue. However, these traditional methods had several limitations. They were time-consuming and labour-intensive, as they required manual feature extraction. In addition to this, they were not very accurate, as they relied on a limited number of hand-crafted features. Therefore, a paradigm shift occurred, marked by the integration of machine learning approaches in voice fatigue research.

B. LEARNING-BASED APPROACHES

The advent of machine learning algorithms introduced a more sophisticated and data-driven methodology, enabling researchers to explore complex patterns and relationships within voice data, thereby enhancing the precision and depth of fatigue identification and classification. The evolution from machine learning to deep learning in fatigue classification marked a significant advancement in capturing voice feature sets. While machine learning techniques demonstrated effectiveness, deep learning's emergence allowed for more nuanced and intricate feature extraction. Deep learning models, particularly neural networks, excel at automatically learning hierarchical representations from raw data, enabling them to discern complex patterns within voice signals. This transition facilitated a more comprehensive understanding of voice features related to fatigue. Authors in [19] proposed continuous prediction of emotion from voice, performed using a prediction-based learning framework. The major objective of this framework is to fully utilize the unique advantages of multiple regression models using bidirectional long short-term memory (bi-LSTM), and support vector regression. The two models were combined in tandem architecture to create a single cascaded framework. Similar

to the previous work, [20] detailed an approach of applying CNN-based Bi-LSTM network architecture on the German SaarbrÃcken Voice Database to classify four categories: paralyzed vocal cords, normal voices, dysphonia and laryngitis.

Authors in [21] investigated the utility of emotive speech in accurately gauging exhaustion levels across various age groups and times of day, eliciting nine distinct emotions. The analysis focused on features such as fundamental frequency, short-time energy, and power spectral density. The study identified consistent emotional states, revealing correlations between emotions like fear, astonishment, hostility, and mental tiredness. The importance of incorporating affective knowledge for contextual speech recognition, considering factors like the speaker's temperament and psychological state is advocated in [22] by proposing a CNN-based model for speech emotion recognition, for continuous speech. Utilizing the Acted Emotional Speech Dynamic Database (AESDD) in Greek, the model surpassed conventional approaches by avoiding the traditional feature selection process. The CNN architecture, demonstrated superior performance in classifying speech samples within this dataset, as validated by qualitative assessments using AESDD.

Researchers in [23] utilized machine learning to estimate heart rates, employing MFCC to capture speech patterns in both spectrum space and temporal changes of spectral properties. A series of experiments have been conducted in [24] to detect ATC fatigue with the implementation of wavelet decomposition and fractal dimension. Also, these results have been compared with SVM which gave an accuracy of 92.82%, which is higher than the state-of-the-art methods.

The study in [25] discusses the effect of using physiological indices combined with the acoustic features of speech. It could determine the levels of fatigue with an accuracy of 94% for unitary voice and 81% for long speech. Researchers in [26] found that a single machine-learning model cannot solve the problem of detecting fatigue. An ensemble learning method based on self-adaption quantum genetic algorithm (SQGA) heterogeneous learning methods is proposed to solve this problem which obtained an accuracy of 98.5% on real civil aviation radio land-air communication. In [7], the researchers have found that certain phones in speech samples have a predictable dependence on tiredness, and accurate phonetic synchronization and recognition are crucial for voice-based fatigue diagnosis. Using a sentiment score, they eliminated less probable word combinations from the ASR result. Also, they performed cepstral analysis to extract 36 MFCC coefficients to understand the fatigue cues. To achieve this, the group developed a separate ASR system for fatigue detection.

C. HYBRID APPROACHES

Despite the increasing interest in deep learning applications for various physiological and behavioural classifications, the literature on voice fatigue classification remains sparse. This scarcity of studies has prompted an exploration of related domains to gather insights and methodologies that could potentially be adapted for voice fatigue analysis. One notable parallel field that has witnessed extensive research is driver fatigue classification, particularly through the analysis of images. Numerous studies have successfully employed deep learning techniques to assess driver fatigue levels based on visual cues and patterns. The existing literature on fatigue classification has predominantly centred around driver fatigue assessment, with a specific emphasis on electroencephalography (EEG)-based classification methods coupled with deep learning techniques. Researchers in [27], [28], [29], [30], and [31] have extensively explored the use of EEG signals to discern fatigue levels, leveraging the rich information embedded in brainwave patterns. While the primary focus of these studies lies in visualizing and interpreting brain activity associated with fatigue, the methodologies employed provide a valuable foundation for the investigation into voice fatigue classification using deep learning. Despite the distinct modalities of EEG signals and voice data, the success of deep learning in decoding complex patterns from physiological signals remains a common thread. In recent years, people have also identified architectures to design models for identifying and predicting fatigue in continuous speech using windowing/segmenting.

In [29], fatigue image samples from car drivers were analyzed using classical deep belief networks (DBN), Bayesian neural networks (BNN), and artificial neural networks (ANN). The research was based on data from 43 participants and demonstrated improved EEG-based driver tiredness classification with the sparse-DBN classifier. Achieving a ROC area of 0.96, the sparse-DBN classifier exhibited 93.9% sensitivity, 92.3% precision, and 93.1% accuracy. Notably, it outperformed ANN, BNN, and DBN classifiers, enhancing accuracy by 13.8%, 9.5%, and 2.5%, respectively. The study utilized a validation-based early stopping strategy to prevent over-fitting. Researchers in [32] introduced an innovative online simulation tool for car driving, utilizing dry electrodes and wireless technology within state-of-the-art VR technology. The approach involved lateral and vertical EOG feature extraction using sample entropy, retrieval of properties from each EEG sub-approximation band, and feature fusion using canonical Pearson correlation. Classification was performed using a relevant vector machine (RVM). The fatigue trials, spanning 90 minutes with 22 volunteers, demonstrated a median accuracy rate of 90.2%, highlighting the potential of entropy assessment using EOG and EEG as an alternative method for detecting driver drowsiness.

The incorporation of sequential models in speech fatigue classification has revolutionized the ability to capture temporal dependencies and intricate patterns within speech signals. Research in [13] and [33] explores hybrid models that combine the strengths of LSTMs and GRUs to further enhance the accuracy and efficiency of speech fatigue classification.

The importance of using deep learning networks such as the bidirectional LSTM was discussed in [34] proposing a context-aware speech-based system for stress detection. Incorporating context led to improvements in overall F1 scores by up to 14% compared to using only speech features. By measuring both speech and physiological data, wearable devices significantly improved classification accuracy. A stress-level detection-based study from continuous speech is conducted in [35] using deep learning models.

The researchers in [36] have proposed an effective approach for EEG-based fatigue detection using a wavelet de-noising method along with a CNN+LSTM model to determine the fatigue of medical staff. Compared to existing models like DNN, CNN, and LSTM resulting in improved classification accuracy. Gao et al. conducted a comprehensive study on vocal fatigue categorization in [37], involving 88 vocally healthy and exhausted individuals. Utilizing sEMG signals and a unique phonatory aerodynamic system for precise voice sample extraction, the study focused on sub-mandibular and infrahyoid muscle groups. The authors improved their machine learning strategy for vocal fatigue identification, achieving an initial accuracy of 38.8%, later enhancing it significantly to 97.8%. Signal normalization and five-fold cross-validation were employed for data augmentation and better validation accuracy. The research in [38] focused on advancing brain-computer interfaces (BCI) for individuals with neurodegenerative conditions and communication challenges through silent speech detection in the EEG of healthy adults. They used 270 volunteers, for capturing silent speech with eight distinct Russian commands. The study demonstrated the ability to create an EEG-based audio input for individuals not originally trained in the neural network classifier, addressing the needs of those with disabilities. The research in [39] closely aligns with this study, wherein an ensemble learning model is introduced by combining facial and voice features for fatigue detection. Facial features were extracted using OpenCV and Dlib libraries, employing M-CNN and E-CNN. Voice features involved extracting MFCC. The fusion of these features through a stacking model resulted in a fatigue detection accuracy of 97% in real-life experiments with ATC. This outperformed both single models (92% best accuracy) and mixed methods (88% best accuracy).

The effectiveness of neural embeddings for vocal fatigue detection is explored in [40]. Features like x-vectors, ECAPA-TDNN, and wav2vec 2.0 embeddings are compared on a spoken English corpus. Results indicate that these embeddings capture changes in vocal characteristics during prolonged speech. SVMs are employed for classification, yielding an accuracy of 81%, 85%, and 82% for x-vectors, ECAPA-TDNN, and wav2vec 2.0 embeddings, respectively. Another hybrid approach was suggested in [41] for fatigue detection, which involved an EEG-based speech signal to train through a 6-layer CNN to extract high-level features. In addition, temporal features are extracted by passing them into a bi-RNN network followed by a classifier for fatigue detection.

D. SUMMARY

Considering the current state-of-the-art literature in fatigue detection and prediction, no single pattern is observed in the speech data for being in a fatigue state. The research studies mentioned above investigated various aspects of speech features and offered support for the use of classifiers like CNN, Bi-LSTM, and other RNNs in different domains, such as emotion identification and driver fatigue detection. However, many of these experiments did not include the detection of fatigue states in continuous speech, particularly in ATC stations. The potential risks associated with a pilot becoming inattentive or misdirecting the air traffic control system can be quite severe. Consequently, there is a crucial need for robust research and experimentation in the domain of fatigue speech monitoring, particularly in the context of ATC stations.

Hence, the main objective of this research is to fill this gap by conducting a study and proposing feature learning-based solutions for fatigue monitoring in continuous speech for ATC stations. EEG-based fatigue studies are invasive and involve equipment-based monitoring to understand intricate patterns of fatigue. The benefit of the noninvasive method of fatigue analysis is achieved through speech-based feature analysis and representation for fatigue studies. Unlike discrete speech, which typically involves a short duration of around 3-4 seconds, continuous speech provides a more natural and dynamic representation of human communication. The increased importance of continuous speech analysis underscores the need for predictive models for the identification of fatigue in real time. To the best of our knowledge, there are no studies that make use of fatigue transition time-stamp prediction from continuous speech analysis of the ATC corpus. In line with these observations, we aim to adapt and extend established methodologies to the unique characteristics of voice signals, contributing to the nascent field of prediction of vocal fatigue.

III. METHODOLOGY

The research in this paper is mainly carried out for fatigue detection in continuous speech. For this purpose, we have taken the Air Traffic Controller Radiotelephony Fatigue Corpus [42] and conducted 2 main studies: fatigue classification using discrete speech samples and prediction using continuous speech. Each of these pipelines has been separately researched. Further, we have created a corpus for continuous speech using the discrete speech samples from the main dataset. In the initial classification task, we built several classification models for two classes - fatigue and non-fatigue. For the prediction task, a continuous speech corpus is first created. Later, in the prediction task, we primarily predict the timestamps where the ATC is transitioning from fatigue to a non-fatigue state or vice-versa. The process flow

can be visualized in Fig. 1. The detailed description of the various parts of this research are described in the following subsections.

A. DATASET DESCRIPTION

The dataset used in this research is the Air Traffic Controller Radiotelephony Fatigue Corpus speech dataset available through IEEE dataport [42]. Speech samples in this dataset have been recorded by real radiotelephony between air traffic controllers (ATCs) and pilots. The corpus is divided into 7 ATC folders viz. ATC1 to ATC7 as shown in Table 1. Each ATC folder consists of two classes - fatigue and non-fatigue. Throughout this paper, the original ATC dataset will be referred to as the discrete speech dataset. The duration of samples varied from 1-30s. Representative samples of fatigue and non-fatigue speech are portrayed in Fig. 2.

B. FATIGUE CLASSIFICATION USING DISCRETE SPEECH

Fig. 1 illustrates the entire pipeline of fatigue detection which is proposed in this research. The first phase of the research involved the discrete speech samples present in the ATC speech corpus as discussed in Table 1. The sequence of operations involved in fatigue detection based on discrete speech samples is described in the following subsections.

1) EXTRACTING FEATURES USING OPENSMILE

The features from each speech sample are extracted using the openSMILE toolkit [43]. The feature set used for classification is INTERSPEECH 2011 Dataset [44]. Once the files are generated, the data is combined into a *.csv file to prepare the feature vectors for pre-processing.

2) FEATURE NORMALIZATION

The feature set extracted from INTERSPEECH 2011 [44] consists of 4600 features. To standardize the data, all the features are rescaled to a mean zero and unit variance distribution using z-score normalization. The standard score of a sample is calculated as in Eqn. 1

$$z = \frac{x - u}{s} \tag{1}$$

where x is the feature value of the sample, u is the mean, and s is the standard deviation of the training samples.

3) DATA BALANCING

As evident from Table 1, the dataset used in this research is highly imbalanced with non-fatigue majority class and fatigue minority class. Without data augmentation, it would lead to a biased classification model which will in turn yield poor performance for fatigue since it forms the minority class distribution. To tackle this problem, we have augmented the data using multiple techniques as described below.

a: SYNTHETIC MINORITY OVERSAMPLING TECHNIQUE (SMOTE)

SMOTE [45] is a type of data augmentation for the minority class. Oversampling the minority class is one way to deal with unbalanced datasets. SMOTE selects similar examples in the feature space and creates a new sample at a location along the line of separation.

b: RANDOM UNDERSAMPLER

Random undersampler selects samples at random, with and/or without replacement, sample down the majority class or classes. This way, we can reduce the samples of the majority class and balance it with the minority class [46].

4) FEATURE FILTRATION TECHNIQUES

To drop unnecessary features, we have used Pearson Correlation. A correlation threshold of 0.815 was selected for the experimentation. Features having a high correlation with other features have been removed because they are redundant.

To prioritize key features, it is essential to extract those with high entropy values. The entropy is calculated using Eqn. 2. For each observed value of a, the dataset may be divided into groups, and the conditional entropy can be determined by adding the ratio of examples in each group out of the whole dataset multiplied by the entropy of each group.

$$H(S|a) = \Sigma S_a(v) S_v \times H(Sa(v)) \tag{2}$$

A collection of samples wherein parameter a has the magnitude of v is represented by the entropy $H(Sa(v))$, where $S_a(v)$ is the proportion of the number of cases in the dataset where parameter a must have the value v. The top 10 features having high entropy values are considered as they contain maximum information.

5) FATIGUE CLASSIFICATION IN DISCRETE SPEECH USING LEARNING-BASED APPROACHES

The study is conducted on six classifiers, viz. K-Nearest Neighbours (KNN), Random Forest (RF), XGBoost, AdaBoost, Histogram GB and 1D-CNN. The train-validation-test split ratio that we followed was 70:20:10. The classifiers used are described in the following subsections.

a: KNN

K-NN is a non-parametric algorithm that classifies the test instance resembling the current classes the most, presuming that the new object and the previous cases are significantly compared. After storing all the previous data, a new piece of information is categorized using the algorithm according to its similarity.

b: BAGGING ALGORITHMS

Bagging hunts out a varied collection of clusters, by changing the training sets. Random forest algorithm is used in this work which aggregates the results from multiple decision trees applied to various selections of the original dataset.

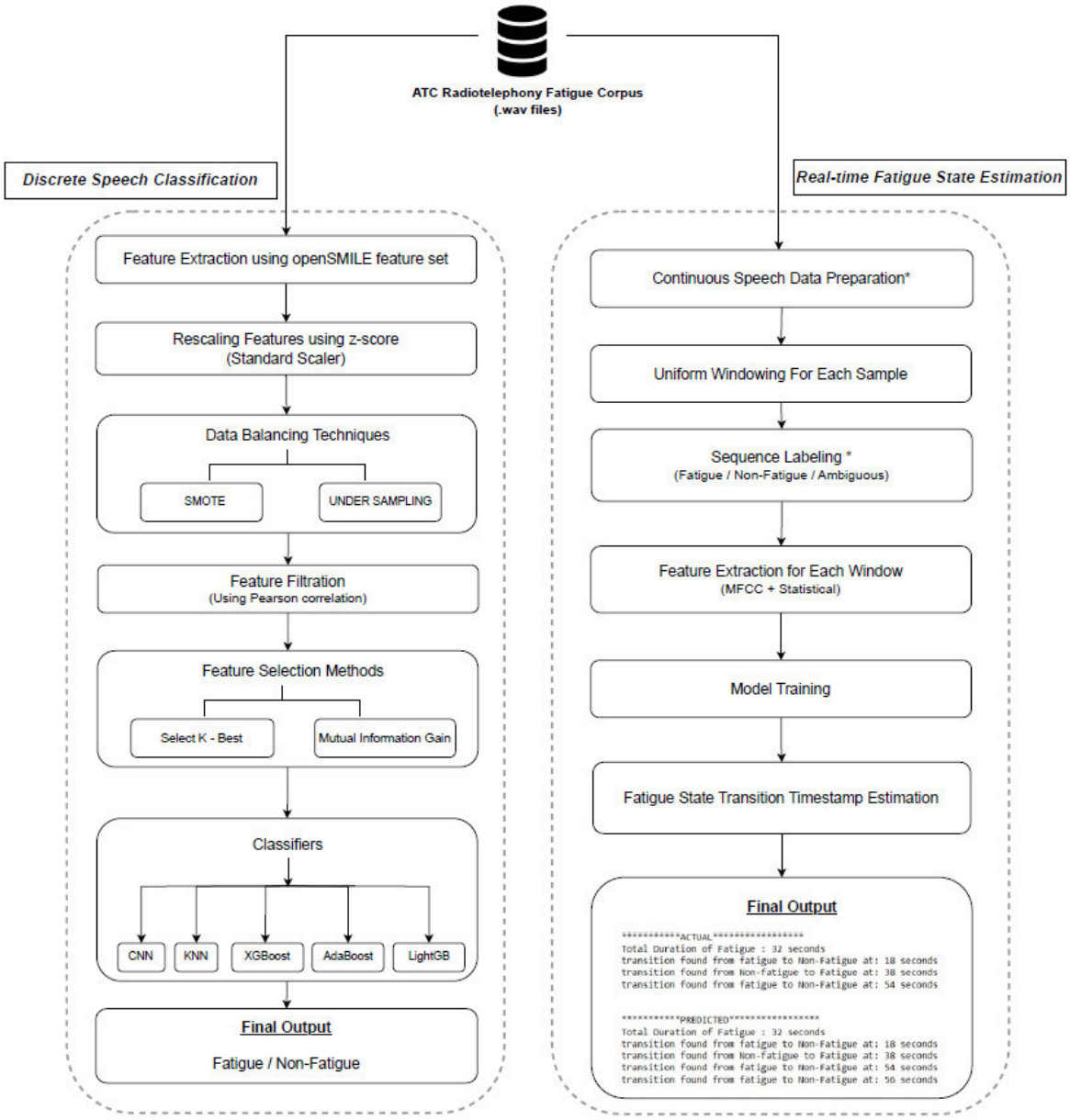


FIGURE 1. Complete System Architecture of the proposed model for ATC fatigue detection.

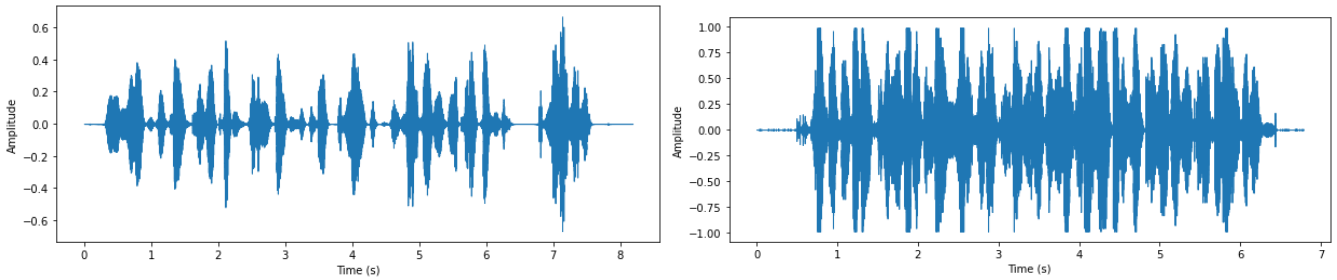
TABLE 1. Statistics of the dataset.

Dataset	Number of Fatigue samples	Number of Non-Fatigue samples	Total
ATC1	415	1182	1597
ATC2	199	1110	1309
ATC3	155	1047	1202
ATC4	164	837	1001
ATC5	152	1421	1573
ATC6	336	1513	1849
ATC7	245	824	1069
Total	1666	7934	**9600**

c: BOOSTING ALGORITHMS

Boosting aims to alter the learning data to draw prominence to instances where past fit classifiers have misfit on the training sets and bias training data in favour of the difficult examples. We have used three boosting algorithms:

1) XGBoost: A simultaneous tree boosting technique that can be used to efficiently solve several data mining problems [47]
2) AdaBoost: Often referred to as Adaptive Boosting, is applied in an ensemble environment with



(a) Fatigue Speech sample

(b) Non-fatigue speech sample

FIGURE 2. **Speech waveform for fatigue & non-fatigue sample from the ATC dataset.**

decision trees having a single level or with a single split [48].

3) Histogram GB: Used to minimize the bias error of the model. Native support for missing values is provided by this estimator. At each split point during training, the tree grower learns, where samples with missing values should be given. If there are no missing values for a particular feature were discovered during training, samples with missing values are mapped to the maximum sampled distribution [49].

d: DEEP LEARNING ALGORITHMS

1) 1D-CNN: The efficiency of utilizing CNNs for sequence classification tasks such as speech is that they may learn from the unprocessed time series data, thereby negating the need for manually designing input characteristics. Once the model has identified meaningful patterns in the raw time series data, its predictions should be comparable to those models that rely on human-engineered features. Though lightweight deep learning models involving time-frequency convolutions are the need of the hour [50], here, we model a 1D-CNN for fatigue detection for better feature mapping and classification.

The discrete speech-based analysis was conducted to understand the features responsible for fatigue classification in ATC. However, to mimic a real-time situation, it is essential to analyze continuous speech data, which is more representative of the typical ATC situation.

C. CONTINUOUS SPEECH CREATION & FATIGUE TIMESTAMP PREDICTION

The transition to continuous speech classification aligns with the broader objective of creating intelligent systems capable of seamless and natural interaction with human operators. The prediction task at hand is to detect the fatigue state in continuous speech. At this stage, we are left with discrete speech samples spanning a range of short durations. So, the first step is to create continuous speech data by combining these discrete samples. The total dataset consists of data from 7 ATCs. For accurate prediction, we considered only discrete samples with a duration greater than 5 seconds. This is because, for discrete samples, the duration varied from 1 second to 30 seconds and the information extracted is not sufficient to predict if the sample is fatigue or non-fatigue. We prepared the continuous speech samples, in such a way that the model becomes speaker-independent while predicting the transition. Every concatenated sample is created speaker-wise. For every set of seven concatenated samples that are created, the data preparation loop gets repeated. This is to ensure that a sufficient number of samples from each ATC are covered to ensure efficient training. Each complete sample will have only one ATC as a speaker in the entire file to ensure accurate training.

Fig. 3 illustrates the proposed method for creating a single continuous speech sample by concatenation of 4 segments, termed chunks. These 4 chunks have been made in such a way that two are from a fatigue combination and two are from a non-fatigue combination. Each chunk is the concatenation of three individual speech samples of the same class. After chunk creation, these are concatenated at random positions to form a long continuous speech sample. The data is created in such a way that every discrete sample is used only once during the chunk preparation. The average duration of each fatigue/non-fatigue sample being concatenated is 5s. In the prediction task, we need to find the timestamps at which there is a transition from either fatigue state to non-fatigue state or vice-versa in the continuous speech samples created. Once the continuous speech is ready, we segment the continuous speech into windows of fixed size. Features extracted for each window are passed into a model as a single feature vector using sequence labelling [51]. As it is continuous speech, we train it using recurrent neural networks (RNNs) which are capable of considering contextual information. These models consider the relationships between previous and current speech samples, allowing for more accurate and coherent predictions. For the prediction task, we classify speech samples into three classes: fatigue, ambiguous and non-fatigue. The output will comprise of predicted class for each window for a continuous speech sample. Here, the ambiguous class indicates the speech sample encompassing both classes - fatigue and non-fatigue. The process overview of the prediction task is shown in Fig 3. Each continuous sample created is segmented into a fixed window of size 2s.

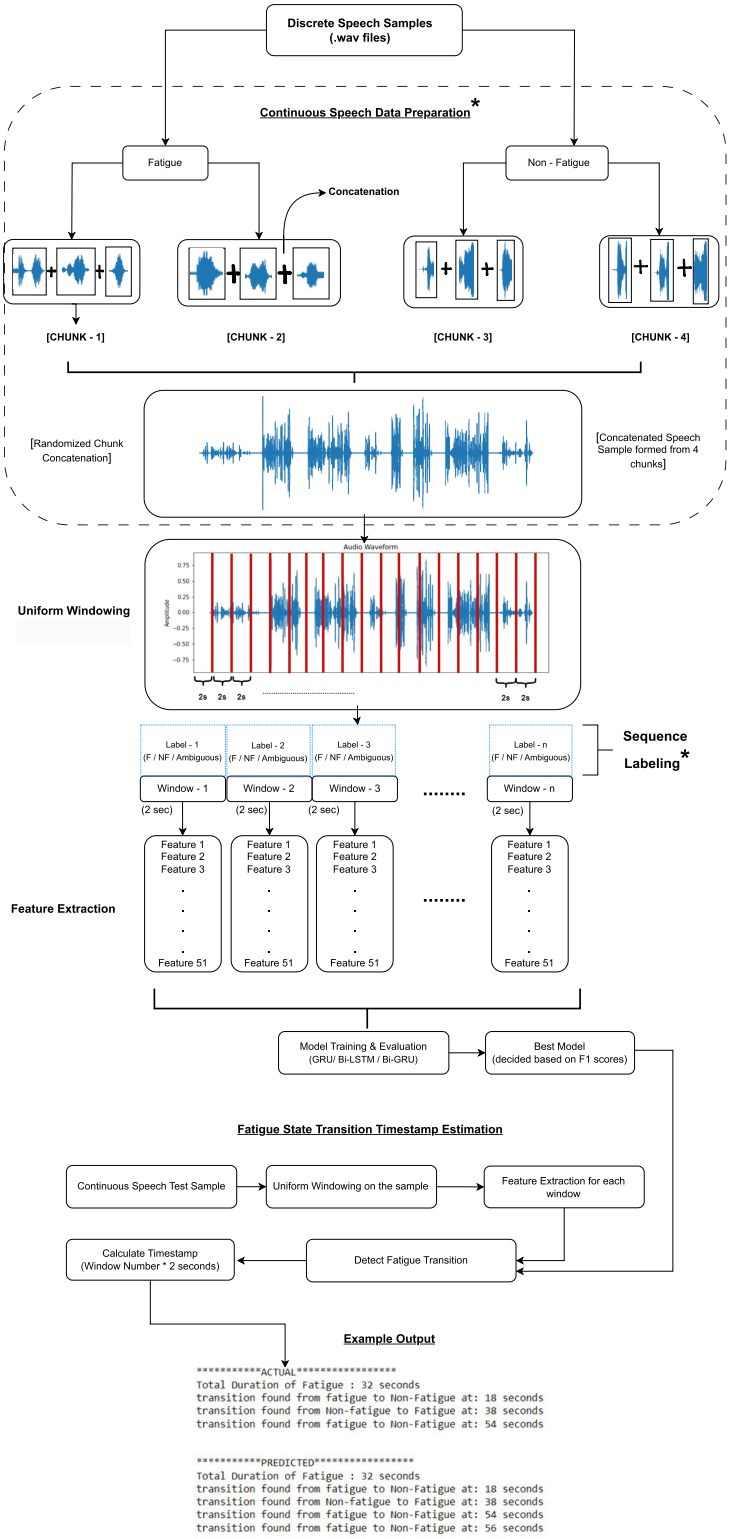


FIGURE 3. Proposed Pipeline for Continuous Speech creation & Fatigue state Prediction.

So, the number of windows (n) generated would be d/2, where d is the duration of the continuous sample. Overall, a set of 18k+ continuous speech samples was created for model training and testing.

1) FEATURE EXTRACTION

The features extracted from discrete speech are not sufficient for the analysis of continuous speech. In continuous speech, the detection of transition from fatigue/non-fatigue also requires statistical features like speech rate, and pitch variation over multiple segments of speech. The use of different feature sets for discrete speech classification and continuous speech prediction in fatigue detection applications can be justified based on the nature of the tasks and the capabilities of the models used. For the task of classifying short, discrete speech samples, the Interspeech 2011 feature set is highly effective. This feature set captures a comprehensive range of acoustic properties and statistical summaries from short segments. Additionally, 1D CNNs typically require fixed-length input vectors. The Interspeech 2011 feature set provides a standardized representation that aligns well with this requirement, facilitating efficient training and accurate classification of discrete samples.

For continuous speech prediction, where the model needs to capture temporal dependencies and long-range patterns, RNNs like LSTM, GRU, or simple RNNs are more appropriate. These models are designed to handle sequences of variable lengths and capture temporal dynamics. MFCCs provide a time-dependent representation of the speech signal, capturing the spectral properties frame by frame. When combined with statistical features over multiple segments, this approach offers a richer and more detailed temporal profile of speech. This sequential representation aligns well with the strengths of RNN in modelling temporal sequences and capturing patterns over time. For short, discrete speech samples, capturing detailed and comprehensive acoustic features using the Interspeech 2011 feature set is effective because the temporal context is limited, and the model can focus on intricate details within each short segment. For continuous speech, capturing temporal dynamics and changes over longer durations is crucial. MFCCs, combined with statistical features, provide a balance between capturing short-term spectral properties and long-term variations, enabling the model to understand and predict fatigue states based on extended speech patterns.

Extracting the extensive Interspeech 2011 feature set for short, discrete samples is computationally feasible and provides rich information for each segment. For longer, continuous speech, the segmentation is carried out at 2s windows each, and we consider the transition between fatigue to non-fatigue states and vice-versa. Correlation analysis of the Interspeech 2011 feature set revealed that the features have less discriminative property to accurately classify the transitions between states, which forms a significant part of this work. Statistical features provide a meaningful representation and discriminative power to classify even ambiguous regions without overwhelming the model with excessively high-dimensional input. In summary, the Interspeech 2011 feature set is well-suited for detailed analysis of short segments, while MFCCs and statistical features provide a balanced and efficient representation for modelling longer temporal sequences in continuous speech.

Building on these observations, a total of 51 features—comprising 39 MFCC coefficients, loudness, and various statistical features, collectively referred to throughout this paper as the 'MFCC+Stat' combination—are used for continuous speech prediction, as detailed in Table 2. MFFCs are known for their compact representation, frequency domain focus, and robustness to noise. They capture phonetic and time-varying information, making it effective for machine learning models. Fatigue often leads to a decrease in overall energy levels, which can be reflected in speech loudness. Individuals under fatigue might speak with reduced loudness, impacting the overall strength and intensity of their speech. Statistical features quantify variability, identify discriminative patterns, provide efficient representations, and aid feature selection.

1) MFCCs are a fundamental feature in speech processing, widely employed for various applications including fatigue detection. MFCCs capture the essential characteristics of a speech signal by converting the power spectrum of the signal into a compact representation. This transformation is particularly valuable because it mimics certain aspects of human auditory perception, enabling better discrimination of important speech features. MFCCs are utilized in the context of fatigue detection in speech to extract relevant information about the vocal features influenced by weariness [52]. Fatigue can cause changes in pitch, tempo, and other auditory qualities of speech patterns. MFCCs excel at capturing these fluctuations because they effectively isolate frequency components vital to human auditory perception while eliminating less important information.
2) Loudness: Perceived volume of sound and is crucial for understanding speech signals, quantified using the Sound Pressure Level (SPL) Eqn. 3

$$SPL = 20 * log \frac{p}{p_{ref}} \tag{3}$$

The loudness of a speech sample is also important in diagnosing fatigue. Monitoring fluctuations in loudness over time can reveal insights regarding a speaker's vocal strain and weariness. When someone is tired, they may speak with less volume, which might result in a fall in SPL. It is possible to detect indicators of exhaustion or vocal fatigue-related diseases by examining the SPL fluctuations in voice samples recorded at different periods.
3) Statistical Features: Important in speech processing because they provide useful insights into the many aspects of longer-duration audio. Table 2 details the statistical features utilised during feature extraction from continuous speech.

D. TRAINING & TESTING

Before passing the extracted features into the model, we perform sequence labelling to assign a label or category

TABLE 2. List of Statistical features extracted from the continuous speech segments.

Metric Name	Description	Computation
Mean (μ)	Average value	$\frac{\Sigma(x)}{n}$
Standard Deviation (σ)	Dispersion of data points from μ	$\sigma = \frac{\sqrt{(\Sigma(xi-\mu)^2)}}{n}$
Maximum Value	Highest feature value	—
Minimum Value	Lowest feature value	—
Median	Mid feature value	—
Skewness	Asymmetry of the data distribution	$\frac{\Sigma(xi-\mu)^3}{(n*\sigma^3)}$
Kurtosis	Peak of the data distribution	$\frac{\Sigma(x_i-\mu)}{(n\times\Sigma^4)}$
1st Quartile	Value below which 25% of the data falls	—
3rd Quartile	Value below which 75% of the data falls	—
Mode	Most frequent value within the dataset	—
Interquartile Range (IQR)	Spread of the middle 50% of the data	—

to each element in a sequence of text. Similarly, we assign each window a class/tag and store them in a list for each continuous sample which act as true labels for training. Further to continuous speech segmentation using 2s windows, some windows are a mix of both fatigue and non-fatigue (transition from one state to another). Features were extracted from these 2-s windows, which will be passed as input data for the classifier. Therefore, we have considered three speech classes for training as depicted in Table 3.

TABLE 3. Segregating the class names into respective labels.

Encoded label	Class label name
0	Ambiguous (Window segment containing a mix of both fatigue and non–fatigue speech)
1	Fatigue
2	Non – Fatigue

Our focus is on the accurate detection of ambiguous parts in the whole speech sample, where the transition from fatigue to non-fatigue or vice-versa happens. Since each sample varies in duration, we need to adjust its duration, so that the dimensions remain the same for all the samples. To resolve this, we followed the technique of padding - each sample is padded with zeros to match the duration of the longest sample. The corresponding class/tag for each of these windows will be "−1", which is ignored by the classifier during training. For training, we used three RNN models - Bi-LSTM, GRU and Bi-GRU. A total of 18908 continuous samples are considered. Out of this, 70 % (13235) is used for training, 15 % (2836) for validation and 15 % (2837) for testing. The output of the model would be a list of arrays containing the predicted class/tag for each window for each continuous sample. Next, we check for the occurrences of ambiguous classes for each sample and extract the timestamp corresponding to that window. This timestamp denotes the transition from fatigue to a non-fatigue state or vice versa. The hyper-parameter settings for the models used are listed in Table 4.

Most of the hyper-parameter settings in Table 4 are the default values used in learning algorithms. The k value for k-NN has been empirically determined by using different values ranging from 1 to $\sqrt{No.of\ datapoints}$. For XGBoost, we additionally used a correlation threshold of 0.7 to reduce collinearity among features. Keeping into account the computational constraints, the number of estimators is fixed at 100 for RF and 50 for Adaboost. In 1-D CNN, Relu activation function is used alongwith a max-pooling size of 5 and stride 2 to reduce spatial dimensions while retaining essential features. Dropout regularization was used to ensure generalisation to new test data. Adam optimizer, capable of handling discrete and continuous data was used to avoid vanishing gradient issues. The deep learning models were also used with default hyper-parameter settings. Softmax activation is used at the output node. The LSTM and dense units were chosen to provide adequate capacity to learn the long-term dependencies in continuous ATC speech while providing a balance between dimensionality reduction and model complexity. The entire sequence of continuous speech creation and fatigue state detection is illustrated as provided in Alg. 1.

E. MODEL EXPLAINABILITY

Since the model is fed with a large amount of data, a detailed idea of the features contributing to the model performance is required. Explainability is brought into the fatigue model using SHAP (Shapley Additive explanations) [53]. SHAP values tell us how much each feature contributes to the model's prediction, considering all possible interactions between features. Shapley value for a feature j is calculated using Eqn. 5.

$$\phi_{j(x)} = \Sigma_S[v(S \bigcup j) - v(S)]/|S| \quad (5)$$

Algorithm 1 Fatigue Detection in Continuous Speech

Require: Three fatigue clips C_{F1}, C_{F2}, C_{F3}, and three non-fatigue clips C_{NF1}, C_{NF2}, C_{NF3}; window length $(L) = 2$ seconds

Combine Clips into Chunks:
  Fatigue Chunk $Chunk_F \leftarrow C_{F1} + C_{F2} + C_{F3}$
  Non-Fatigue Chunk $Chunk_{NF} \leftarrow C_{NF1} + C_{NF2} + C_{NF3}$
Create Final Speech Sample:
  Concatenate two chunks of $Chunk_F$, say $Chunk_{F1}$ and $Chunk_{F2}$ with two chunks of $Chunk_{NF}$, viz. $Chunk_{NF1}$ and $Chunk_{NF2}$ to create final sample S
  $S \leftarrow [Chunk_{F1} + Chunk_{F2} + Chunk_{NF1} + Chunk_{NF2}]$

Segment Final Sample into Windows:
  Divide S into 2-second segments (windows of length L) to obtain segments $W_1, W_2, \ldots, W_n$
Feature Extraction and Tagging:
for each window W_i **do**
  Extract feature vector F_i (MFCC + Statistical features)
  Assign original label to W_i: F (fatigue), NF (non-fatigue), or A (ambiguous, if applicable)
end for
Train Model:
Use features F_i and labels to train a sequence labeling model to classify W_i as F, NF, or A
Prediction and Transition Detection:
For a test sample, segment it into 2-second windows and predict labels for each window
Identify timestamps where transitions occur between fatigue and non-fatigue states based on label changes
return Transition timestamps between fatigue and non-fatigue states

$$Timestamp = (Window\ index * 2s) \tag{4}$$

where:

1) $\phi_{j(x)}$ is the Shapley value of feature j in the context x.
2) S represents subsets of features excluding j.
3) N denotes the set of all features.
4) $v(S)$ is the value of the model prediction when only features in subset S are considered.
5) $v(S \bigcup j)$ is the value of the model prediction when features in subset S are considered along with feature j.
6) $|S|$ denotes the number of elements in subset S.

There are two main steps to calculate SHAP values:

1) Data masking: For each feature, we create a set of background data points by masking out that feature. This means that we set the value of that feature to its reference value.
2) Local explainer: For each data point, we use a local explainer to estimate the model's prediction for the masked data points.

The SHAP value for a feature is then calculated as the average difference between the model's prediction for the original data point and masked data points. This difference represents the marginal contribution of that feature to the model's prediction. A positive SHAP value for a feature means that the feature increased the model's prediction while a negative SHAP value represents that the feature decreased the model's prediction. The absolute magnitude of the SHAP value indicates the importance of the feature.

F. PERFORMANCE EVALUATION METRICS

Performance measures used in this research are recall, precision, F-measure, and accuracy to evaluate the system [54]. Table 5 gives a brief overview of these metrics.

IV. RESULTS & DISCUSSION

Three major components of this study, each addressing a distinct aspect of speech fatigue classification, are as follows:

1) ATC Fatigue classification from Discrete speech
2) ATC Fatigue and transition state detection from Continuous speech
3) Testing Fatigue Prediction on Continuous Speech Data

In the first subsection, we present the outcomes of various classification models applied to the discrete data classification task where we have tested our classifiers on different combinations of data after oversampling (SMOTE) and undersampling. The speech samples are categorized into two distinct classes: fatigue and non-fatigue. In addition to various shallow models, a deep 1D-CNN was also utilized for discrete data classification, to understand the usability of deep models in fatigue data classification. In the second section, we examine continuous data categorization and evaluate how well the models perform in predicting fatigue on a continuous scale. We have used GRU, Bi-GRU, and Bi-LSTM for classification which allows us to capture and process the dependency in continuous data because RNNs have the capability of sequence labelling tasks. With this method, we can investigate the models' capacity to detect transitions between fatigue and non-fatigue states, which is referred to as the "Ambiguous" tag. All the considered machine learning and deep learning pipelines have been implemented using Jupyter Notebook (Anaconda Environment, Python 3.8) installed on a system with intel i7 8th generation chipset, 16 GB of DDR4 RAM and GPU of Nvidia MX 250 with 2GB DDR5.

A. ATC FATIGUE CLASSIFICATION FROM DISCRETE SPEECH

The first stage in classification involves the selection of top-10 features from the Interspeech 2011 feature set. These features were selected based on mutual information gain and the Pearson correlation coefficient computed on the dataset

TABLE 4. Hyper-parameter settings for fatigue prediction models using discrete & continuous speech.

Type of Data	Model	Parameter	Value
Discrete	KNN	Neighbors	9
	XGBoost	Loss Function	Multi-class log-loss
		Correlation threshold	0.7
	Random Forest	Estimators	100
	AdaBoost	Estimators	50
		Learning Rate	1
	LightGBM	Boosting Type	Gradient-boosted decision trees
		Learning Rate	0.1
	HistogramGB	Loss function	Log Loss
		Learning Rate	0.1
	1D-CNN	Neurons	64
		Kernel size	5
		Stride	1
		Padding	same
		Activation Function	Relu
		Max-pooling Pool size	5
		Max Pooling-Stride	2
		Dropout Units	0.2
		Dense Layer Units	512
		Dense Activation Function	relu
		Dense Dropout units	0.3
		Dense Optimizer	Adam
Continuous	GRU/bi-GRU	GRU units	128
		Dense Units	64
		Dense Activation Function	Relu
		Output Activation Function	Softmax
		Optimizer	Adam
		Loss function	Categorical crossentropy
	Bi-LSTM	LSTM units	128
		Dense Units	64
		Dense layer Activation Function	Relu
		Output Layer Activation Function	Softmax
		Optimizer	Adam
		Loss function	Categorical crossentropy

TABLE 5. Evaluation Metrics employed to assess the models.

Metric	Computation
Accuracy	$\frac{Correctly\ recognized\ test\ samples}{Total\ number\ of\ test\ samples}$
Recall	$\frac{True\ Positives}{True\ Positives+False\ Negatives}$
Precision	$\frac{True\ Positives}{True\ Positives+False\ Positives}$
F1-Score	$\frac{2\times Recall\times Precision}{Recall+Precision}$

TABLE 6. Selected best features from discrete speech.

Sampling Technique	Top features
SMOTE	Audio spectral feature Pulse Code Modulation feature Fast Fourier Transform RMS energy Zero Crossing Rate
Under sampling	Audio spectral feature Fast Fourier Transform Fundamental frequency
Without Sampling	Audio spectral features Pulse Code Modulation – RMS energy Pulse Code Modulation – Fast Fourier Transform Fundamental frequency

after feature selection. The best features selected for each category of sampling are listed in Table 6.

Further to top-10 feature selection using the correlation and mutual information, SHAP tool analysis has been done on the best-performing model, in this case, the XGBoost model for discrete speech. The SHAP explainer used here is the Tree explainer which uses the structure of the tree to define feature perturbations. The SHAP tool effectively elucidates the importance of each feature in the dataset by attributing a Shapley value, which represents the contribution of that feature to the model's predictions. Fig. 4 represents

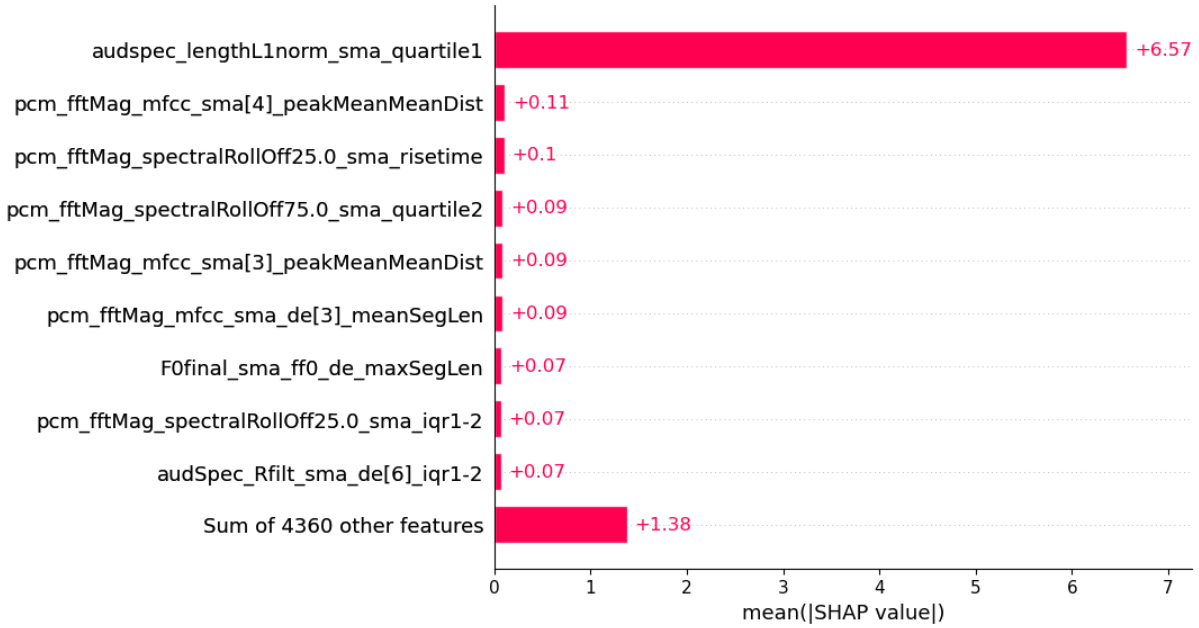


FIGURE 4. SHAP tool analysis of the top features for discrete speech classification.

the contribution of the top 10 features using the SHAP tool, and Table 6 describes the top features in each of the SMOTE and undersampling methods of discrete data. The top 10 features selected in Table 6 are a result of feature selection using Pearson correlation and Mutual Information Score.

The complexity of feature interactions and the hierarchical nature of the features learned by CNNs pose significant challenges for the SHAP tool, which is designed to handle more straightforward feature contributions typically found in classical machine learning models. Consequently, applying SHAP analysis to a CNN would not yield meaningful or interpretable results, as the features are not explicitly defined but rather emergent properties learned during the training phase. Observing all the features that were selected, it can be seen from Fig. 4 and Table 6 that the top features are essentially the same with audio spectral length appearing as the most important feature. This feature is a part of auditory spectral features corresponding to the loudness of the audio dominating all other features present. The convolution layer already has a built-in feature selection technique. Therefore, the CNN model was able to classify the data better than the machine learning classifiers. The 1D-CNN did not give better results in the absence of feature selection especially for under-sampled data.

Table 7 focuses on the F1 scores of the proposed models, which provide essential insights into their ability to accurately differentiate between speech samples with and without fatigue. These results offer valuable information on the models' effectiveness in accurately identifying instances of fatigue and non-fatigue, contributing to a better understanding of their classification performance. The classifiers are KNN, XGBoost, Random Forest (RF), Ada Boost, LightGBM, and Histogram GB. The goal of these classification tasks is to predict whether a sample belongs to the ''Fatigue'' or ''Non-Fatigue'' class. The table lists the results of applying three different techniques to the data: no-sampling under-sampling, and over-sampling. The results are presented in terms of accuracy, precision, recall and F1 score for both the ''fatigue'' (F1 Score (Fatigue)) and ''non-fatigue'' (F1 Score (Non-Fatigue)) classes.

From Table 7, it is observed that the KNN classifier struggled to accurately identify instances of fatigue, regardless of the data sampling technique employed. The imbalanced class distribution poses a challenge for the model, leading to sub-optimal performance in detecting fatigue instances. The boosting algorithms demonstrated exceptional performance across all data sampling techniques. Under both the original data and under-sampling cases, the classifier achieves perfect accuracy and F1 scores for both the fatigue and non-fatigue classes. The over-sampling approach results in slightly reduced accuracy, but the F1 scores remain exceptionally high, indicating the classifier's effectiveness in accurately distinguishing between fatigue and non-fatigue instances. The robustness of boosting algorithms to class imbalance and its consistently high accuracy make it a strong candidate for speech fatigue classification tasks. RF treats all classes equally, so the majority class often dominates the training process, leading to bias toward it. The majority voting across trees typically favours the majority class, which can overshadow predictions for the minority class. Moreover, since RF does not iteratively focus on correcting errors,

TABLE 7. Performance evaluation on discrete speech data using the top features.

Classifier	Sampling		Precision		Recall		F score		Average
	Under	Over	Fatigue	Non-Fatigue	Fatigue	Non-Fatigue	Fatigue	Non-Fatigue	Accuracy
KNN	-	-	53	86	29	95	37	90	83
	-	✓	30	93	80	61	45	74	64
	✓	-	23	98	31	97	37	48	43
XGBoost	-	-	100	100	100	100	100	100	100
	-	✓	100	100	100	100	100	100	100
	✓	-	100	100	100	100	100	100	100
Random Forest	-	-	55	87	35	94	43	90	84
	-	✓	54	96	85	85	66	90	85
	✓	-	57	88	93	42	91	48	84
AdaBoost	-	-	100	100	100	100	100	100	100
	-	✓	100	100	100	100	100	100	100
	✓	-	100	100	100	100	100	100	100
LightGBM	-	-	100	100	100	100	100	100	100
	-	✓	100	100	100	100	100	100	100
	✓	-	100	100	100	100	100	100	100
HistogramGB	-	-	100	100	100	100	100	100	100
	-	✓	100	100	100	100	100	100	100
	✓	-	100	100	100	100	100	100	100
1D-CNN	-	-	99	99	97	100	100	98	99
	✓	-	98	98	98	98	98	98	98
	-	✓	89	99	99	88	93	94	94

TABLE 8. Comparison with state-of-the-art for discrete speech fatigue classification.

Author (Year)	Feature set	Size of feature set	Classifier	Accuracy (%)
Shen et al. (2021) [14]	INTERSPEECH 2011 Spectrogram-based	4368	Autoencoder + Multiple kernel Learning with SVM	98.35
Nan Wu et al. (2022) [26]	MFCCs	150	SQGA based Ensemble learning	98.50
Proposed work	INTERSPEECH 2011 + Feature Selection	top 10	XGBoost, Adaboost	100
Proposed work	INTERSPEECH 2011 + Feature Selection	top 10	LightGBM	100
Proposed work	INTERSPEECH 2011 + Feature Selection	top 10	HistogramGB	100

it lacks mechanisms to give extra attention to minority class instances.

Additionally, recall for fatigue class is drastically improved for every classifier with the use of the SMOTE but no change in the precision values. Among ML classifiers, precision and recall showed a better result when the undersampling method was used. The sampling method also depends on how the model deals with the variability of data. The RF classifier's specific results with different data sampling techniques can be attributed to the dataset imbalance and the algorithm's inherent randomness. Under-sampling improved the classifier's ability to identify fatigue instances, resulting in better performance for the same class. However, over-sampling might have introduced overfitting and affected overall accuracy. The classifier's focus on fatigue instances may have slightly impacted its performance in the non-fatigue class.

Finally, the proposed model is compared with results from the state-of-the-art models trained on the same dataset. Table 8 gives details on this comparison. For direct comparison, we considered only fatigue detection research utilising the same dataset. Accuracy is considered the metric for comparison. From Table 8, we can observe that the bi-LSTM-based model for continuous speech fatigue detection provides superior results compared to the existing pipelines. Table 8 summarizes the standard research on fatigue detection, classification and prediction. Compared to the current systems and procedures, the combination of ensemble machine learning classifiers gave 100% accuracy for fatigue detection using the reduced INTERSPEECH 2011 feature set after data augmentation and feature selection using mutual information gain.

As mentioned in the methodology, the discrete-time speech sample data was further used to create continuous data and results obtained using this continuous data were discussed.

B. ATC FATIGUE AND TRANSITION STATE DETECTION FROM CONTINUOUS SPEECH

This study aimed to identify the transition in real-time speech from ATCs using multiple feature sets with combinations of

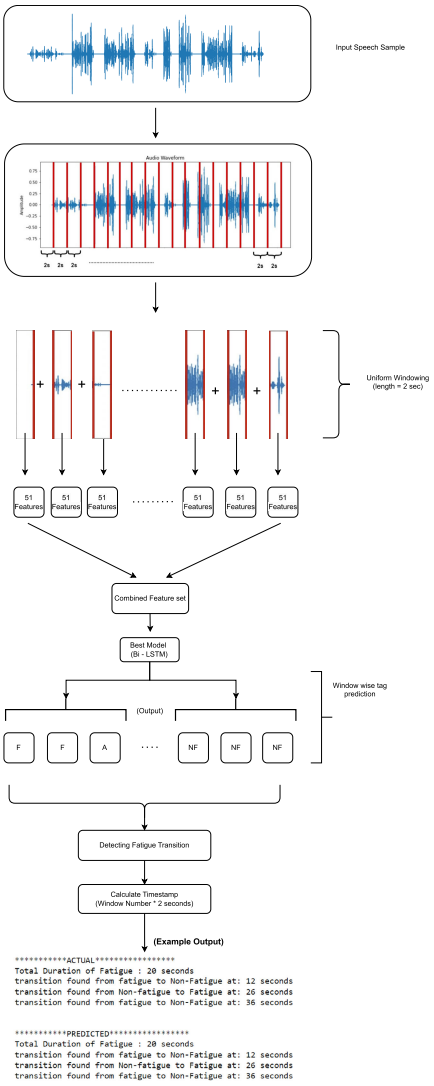


FIGURE 5. Testing pipeline for a continuous speech sample depicting state prediction and timestamp calculation.

MFCC and statistical features. The discrete speech samples are utilised to create continuous speech data and used for the prediction of fatigue, non-fatigue and ambiguous states. Fig. 5 takes a sample of continuous speech and illustrates the process of transition state prediction and generation of timestamps.

Fig. 6 represents the average performance scores for three different models (GRU, Bi-GRU, Bi-LSTM) on the state detection problem with three classes: ambiguous, fatigue, and non-fatigue. The GRU model achieves an average F1 score of 80%, precision of 83% and recall of 78% using the MFCC feature set. Bi-LSTM performed best using the same feature set, with corresponding average values of over 97% in F score, precision and recall. For statistical features also, the same trend followed, with bi-LSTM recording average scores above 83% for all performance metrics. Further, when we use the augmented feature set, the scores for the bi-LSTM model touch above 97% for precision, recall and F score while we obtained an accuracy of about 99% using the same feature set. To further analyse the class-wise performance, we have calculated the values across each class and the same has been tabulated in Table 9 for all the feature-model combinations.

From Table 9, some important observations are made. GRU model performs poorly in detecting ambiguous classes. Even for the augmented feature combination, the average F score recorded for A-class is 35%. However, it performs reasonably well for the fatigue and non-fatigue classes, with F1 scores of 94% and 98%, respectively. The Bi-GRU model shows improved performance compared to the GRU model, achieving a maximum F1 score of 77% for ambiguous classes when using only the MFCC feature set. It also maintains high F1 scores for the fatigue and non-fatigue classes, with values of 98% and 99%, respectively. The bidirectional processing in the Bi-GRU model, which considers the input sequence in both forward and backward directions, helps capture contextual information effectively and contributes to its improved performance. It is interesting to note that for the bi-GRU model, MFCC features give better performance in terms of precision (82%), recall (73%) and F score (77%). On the other hand, the Bi-LSTM model demonstrates the highest performance score for the ambiguous class, with a value of about 94% across all indices. It also performs exceptionally well for the fatigue class, achieving an F1 score of 99%, indicating its ability to accurately identify instances of fatigue. Moreover, the Bi-LSTM model achieves a perfect F1 score of 100% for the non-fatigue class. These results suggest that the Bi-LSTM model excels in distinguishing between non-fatigue and fatigue classes. Furthermore, it demonstrates strong performance in capturing the nuances of transitions between fatigue and non-fatigue states, as indicated by the high F1 score for the ambiguous class. The recurrent architecture and ability to learn temporal dependencies of the Bi-LSTM model contribute to its success in handling complex sequential data such as ATC speech samples.

In summary, for MFCC, the Bi-LSTM model stands out with excellent performance across all three classes. It accurately identifies instances of fatigue and demonstrates a high F1 score for the Ambiguous class, indicating its ability to handle the challenges associated with transitions between fatigue and non-fatigue states. The Bi-GRU model also performs well, with improved F1 scores for the Ambiguous class compared to the GRU model. However, further improvements may be necessary to address the challenge of accurately identifying instances of fatigue and differentiating them from ambiguous transitions. Fig. 7 represents the trend of training, validation loss and accuracy with MFCC.

Secondly, the statistical feature set provided in Table 2 was used to train and test the models. About class-wise results, the GRU model obtained an ambiguous class F1 score of 60% with 76% precision, showing a decent ability to recognize instances of changes from fatigue to non-fatigue states. It performed well for both fatigue and non-fatigue classes, earning an F1 score of 97%, and 99% respectively. The bi-GRU model performs worse than the GRU model,

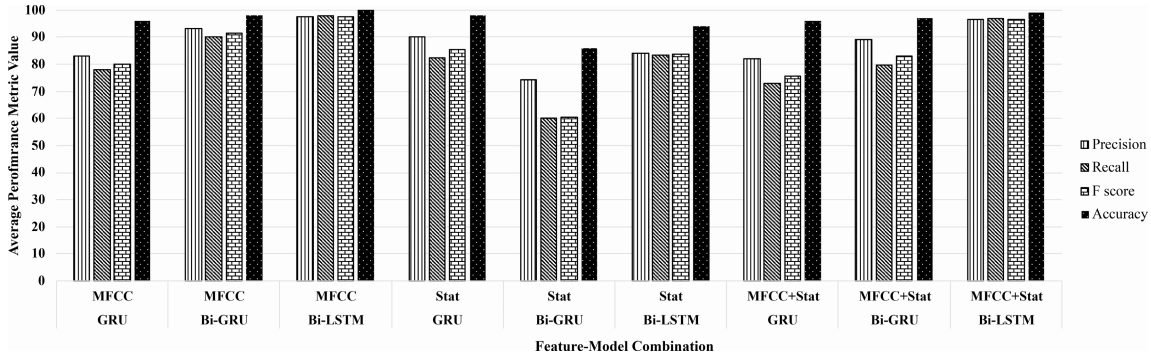


FIGURE 6. Performance Evaluation results using different deep learning classifiers. In each case, the average values across the three classes viz. fatigue, non-fatigue and ambiguous have been taken and plotted.

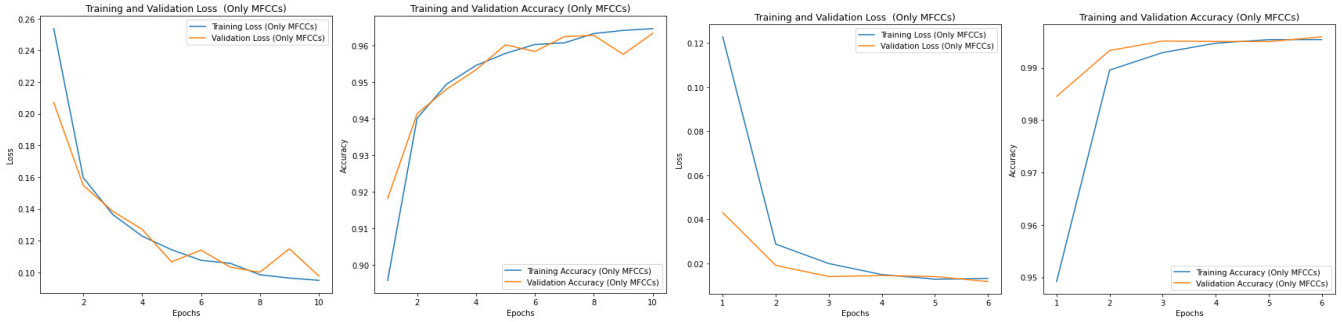
TABLE 9. Performance Scores of individual and combined feature model combinations on continuous speech data.

Feature-Model Combination	Precision			Recall			F score		
	Ambiguous	Fatigue	Non-Fatigue	Ambiguous	Fatigue	Non-Fatigue	Ambiguous	Fatigue	Non-Fatigue
MFCC-GRU	59	91	99	38	98	98	46	95	99
MFCC-Bi-GRU	82	98	99	73	97	100	77	98	99
MFCC-Bi-LSTM	94	99	100	95	99	100	94	99	100
Stat-GRU	76	95	99	49	99	99	60	97	99
Stat-Bi-GRU	64	66	93	10	79	91	17	72	92
Stat-Bi-LSTM	68	87	97	64	89	97	66	88	97
MFCC+Stat-GRU	56	92	98	25	95	99	35	94	98
MFCC+Stat-Bi-GRU	75	94	98	43	97	99	55	95	99
MFCC+Stat-Bi-LSTM	**91**	**99**	**100**	**92**	**99**	**100**	**91**	**99**	**100**

with an F1 score of 17% with recall scores of 10% for the Ambiguous class, suggesting that the model has difficulty in correctly identifying transitions. However, it performs satisfactorily for both fatigue and non-fatigue classes with F scores of 72% and 92% respectively. The Bi-LSTM model performed moderately well at identifying ambiguous states, achieving an F1 score of 66%, which suggests it can only moderately capture the transition between fatigue and non-fatigue. However, the model performed well in classifying fatigue (F1 score of 88%) and non-fatigue (F1 score of 97%). These findings suggest that the feature set selection significantly affects the models' performance. The Bi-LSTM model outperforms the other models for the ambiguous class when employing a statistical feature set, demonstrating its capacity to capture the complexity of ambiguous transitions. The lower F1 score, however, suggests that it still has some difficulty in correctly identifying episodes of weariness. The Bi-GRU model has trouble locating instances of ambiguous transitions, but the GRU model performs well for both fatigue and non-fatigue classes. Fig. 8 represents the trend of training, validation loss and accuracy when used with statistical features as the sole feature set.

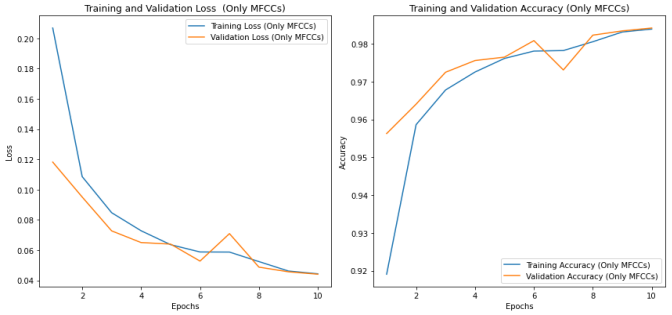
Finally, an augmented feature set of MFCC and the statistical feature set were used to train and test the models. The GRU model had difficulty identifying changes from fatigue to non-fatigue states, with a low F1 score of 35% for the ambiguous class, with a lower recall of 25% for the same. However, it performed well for the other two classes, achieving F1 scores of 94% and 98% for fatigue and non-fatigue, respectively. The Bi-GRU model recorded a slightly better F1 score of 55% for ambiguous class. It also performed well in the other two classes, with precision and F scores above 94% for fatigue and non-fatigue, respectively. The Bi-LSTM model excels at identifying transitions from fatigue to non-fatigue states, providing impressive scores of above 91% for the ambiguous class. Moreover, it demonstrated exceptional accuracy in classifying clear-cut cases, scoring a perfect 100% for the non-fatigue class and an outstanding 99% F score for the fatigue class. Moreover, 100% precision and recall were observed using the augmented feature combination when used in combination with bi-LSTM model. The values are predominantly higher for the non-fatigue class due to the inherent imbalance in the data.

These findings imply that MFCC and statistical feature set work together to enhance performance in capturing the subtle differences across classes. With high F1 scores for all three classes, the Bi-LSTM model emerges as the top performer. It handles confusing transitions well and distinguishes between cases of weariness and non-fatigue with excellent accuracy. When comparing the three feature set combinations



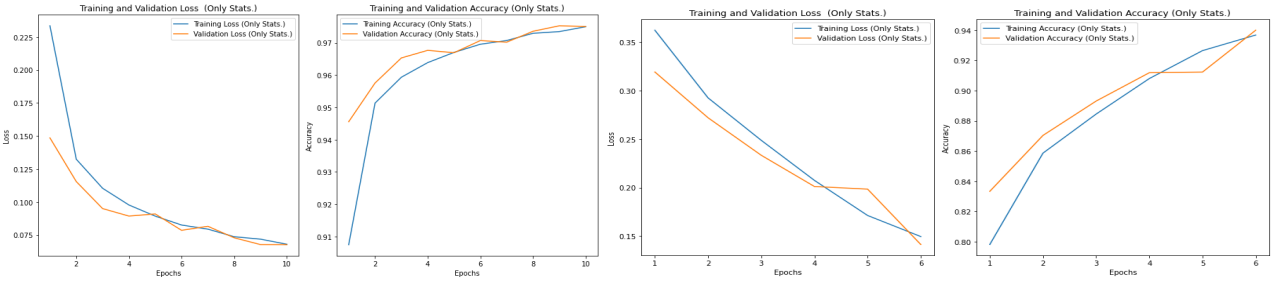
(a) Using GRU model

(b) Using bi-GRU model



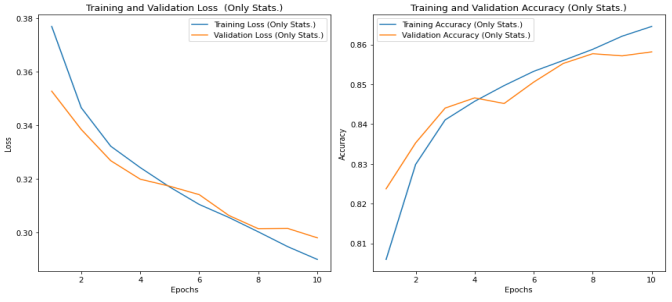
(c) Using bi-LSTM model

FIGURE 7. **Model Validation performance using only MFCC features extracted from continuous speech.**



(a) Using GRU model

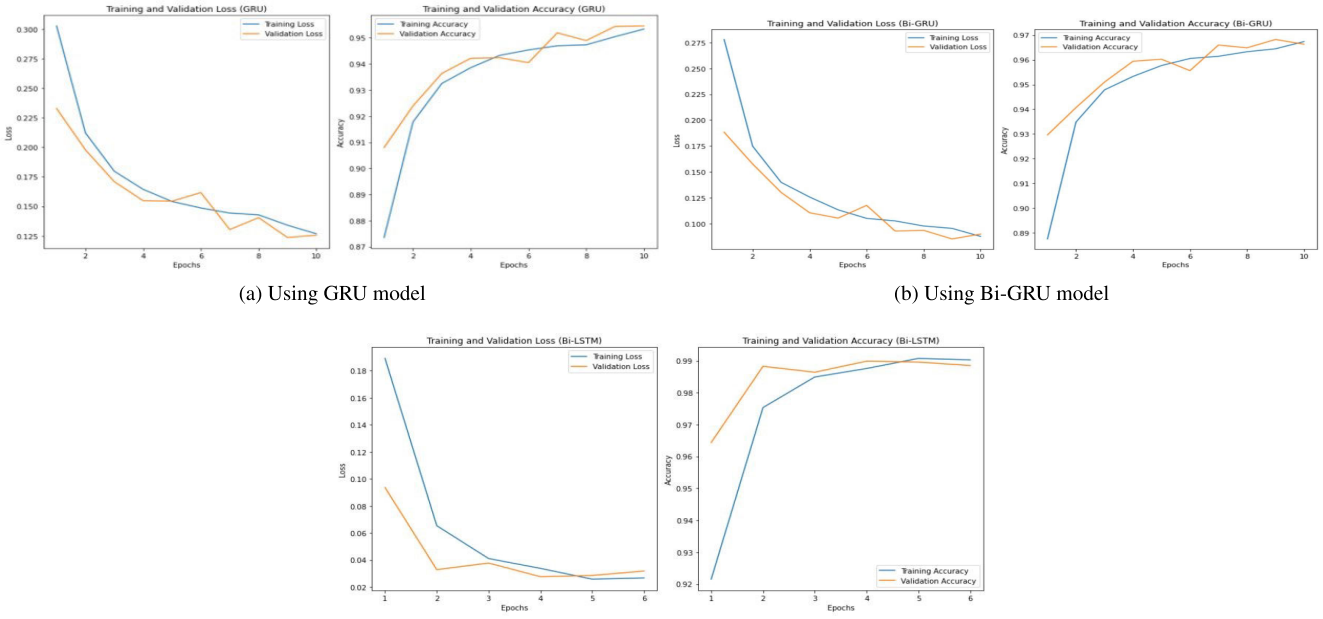
(b) Using bi-GRU model



(c) Using bi-LSTM model

FIGURE 8. **Model Validation performance using only statistical features extracted from continuous speech.**

in the context of their impact on the performance of the models, some notable patterns emerge. Firstly, when using MFCCs as the sole feature set, we observe mixed results. The GRU model demonstrates decent performance for the fatigue and non-fatigue classes. However, it struggles to accurately identify instances of transitions between fatigue and non-fatigue states. While the Bi-GRU model exhibits improvement over the baseline, it still faces challenges



(a) Using GRU model

(b) Using Bi-GRU model

(c) Using Bi-LSTM model

FIGURE 9. Model Validation performance using augmented feature set of MFCC + Statistical features extracted from continuous speech.

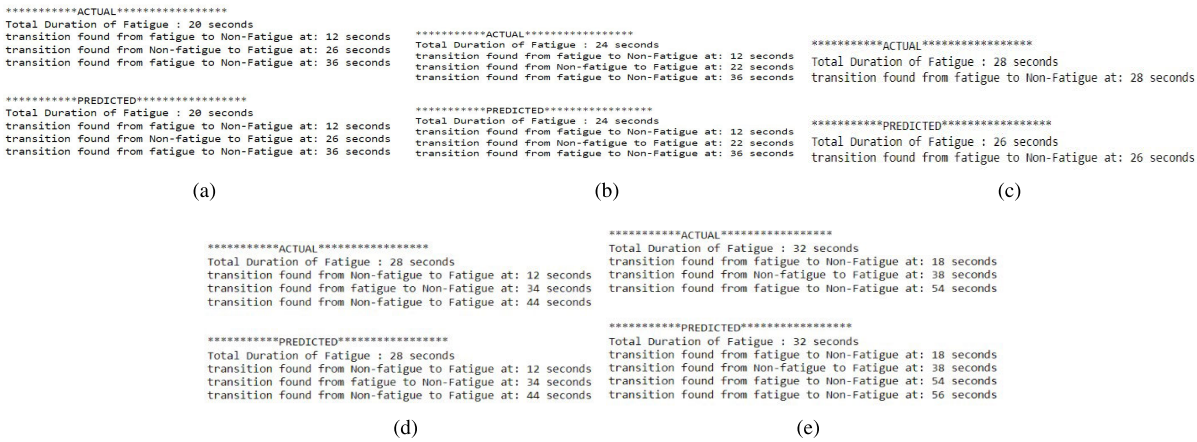


FIGURE 10. Prediction results on continuous data using Bi-LSTM.

in accurately classifying ambiguous transitions. Similarly, the Bi-LSTM model demonstrates proficiency in transition detection and fatigue identification.

In contrast, when using statistical features alone, the GRU model shows better performance for the ambiguous class. The Bi-GRU model, however, still struggles to capture ambiguous transitions accurately. The Bi-LSTM model continues to have difficulties in accurately identifying instances of transition (ambiguous class) but performs well in accurately identifying instances of fatigue. The combination of both MFCCs and statistical features yields the most promising results. Even though the GRU and Bi-GRU models demonstrate lower performance in capturing ambiguous transitions compared to their counterparts using individual feature sets, the Bi-LSTM model stands out as the top performer across all feature set combinations, excelling in accurately identifying instances of transitions, fatigue, and non-fatigue. Fig. 9 represents the trend of training, validation loss and accuracy when used with the complete feature set.

C. TESTING FATIGUE PREDICTION ON CONTINUOUS SPEECH DATA

As observed from Fig. 6 and Table 9, bi-LSTM gave a superior performance on continuous datasets. The complete testing pipeline for one sample of continuous speech is illustrated in Fig. 5 For the real-time fatigue prediction and testing, the bi-LSTM model is subjected to continuous speech data, randomly selected from test data, and the sample test results are represented in Fig. 10.

As seen in Fig. 10, the model correctly identified shifts from fatigue to non-fatigue at 12 and 36 seconds, and from non-fatigue to fatigue at 22 seconds in continuous speech of duration around 50s. The fact that the predicted and ground truths are identical indicates the accuracy by which the model captured these transitions. The success of the Bi-LSTM model for continuous speech fatigue prediction, in this case, shows that it can spot patterns in continuous speech data connected to shifts in fatigue states. A similar kind of understanding can also be inferred from all the five test cases as demonstrated in Fig. 10.

In summary, our proposed methods demonstrate superior effectiveness in fatigue detection, achieving an accuracy of over 94% using MFCCs, combined with statistical features. The optimized feature set and lightweight model identified for both discrete and continuous speech-based fatigue detection holds significant potential for efficient hardware integration. Moreover, the concept of creating a continuous speech dataset and predicting timestamps for fatigue, non-fatigue, and transitional states presents promising novel directions for future research and development in real-time fatigue detection and monitoring for ATCs.

V. CONCLUSION & FUTURE SCOPE

In this research, we have proposed and implemented a fatigue speech detection and state transition prediction system based on various learning classifiers. The first stage of research used discrete speech samples from the ATC radiotelephony corpus and created a reduced feature set for fatigue detection. Following this, a continuous speech dataset of 18000+ samples was created and utilised for fatigue state detection using various deep learning classifiers. Finally, after parameter tuning and detailed analyses, we conclude that the bi-LSTM model outperformed the GRU and Bi-GRU models, by a considerable margin of almost 60%, using simple MFCC and statistical-based features. Additionally, we formulated a comparative study out of this, to gain insightful inferences. In summary, the implementation of deep learning-based fatigue monitoring in continuous speech for ATC Stations holds immense significance in ensuring the safety and efficiency of air traffic management systems. This research provides an innovative and non-intrusive strategy to address the critical issue of fatigue-related performance loss by utilizing state-of-the-art learning models to monitor and detect indicators of exhaustion in real-time from the controllers' speech patterns. The reduced feature set identified during this work is particularly helpful for hardware implementation and prototyping of the fatigue detection system in real-time.

A more secure and efficient worldwide aviation network could result from this system's potential to deliver prompt alarms and permit proactive measures to reduce fatigue-induced mishaps. As a part of future enhancement and extension to the current set of work, we plan to scale the models developed to a multilingual diverse set of ATCs, to design and formulate the problem into a multi-lingual based fatigue monitoring in continuous speech to make it more robust. The models used in this work can be fine-tuned to develop a lightweight application which can be made into a wearable device on the pilot for real-time monitoring of fatigue.

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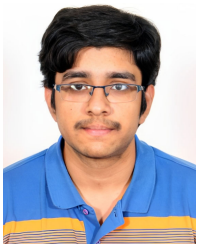
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